FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 10-11-2019

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1 A force $F(x, y, z) = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$ is conservative if, and only if, the following relations are satisfied:

(A) $\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}$

(B) $\frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}$

(C) $\frac{\partial F_z}{\partial x} = \frac{\partial F_x}{\partial z}$

(D) Work done by the force on any closed path is zero.

2. The two endpoints of an 8 m long weightless thread are fixed at the same height at a distance of 4 m. An object of mass 10 kg is hung onto the thread and can move without friction. A horizontal force of magnitude F is exerted on the object to keep it at rest on the thread exactly below one of the endpoints of the thread? If F is removed, the object has an initial acceleration 'a' and eventually acquires a maximum speed v_{max}.

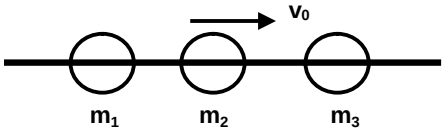
(A) F = 50N

(B) $a = 4m/s^2$

(C) $v_{max} = \sqrt{20\left(2\sqrt{3} - 3\right)}$ m/s

(D) The radius of curvature of path traced by the particle after removal of F will first decrease and then increase.

3. Three balls of mass m_1, m_2 and m_3 can slide without friction along a horizontal rod. (There are holes in the balls.) At a certain moment the middle ball is given an initial speed of v_0 = 5 m/s towards the right. If all collisions are totally elastic pick the correct statement/s .



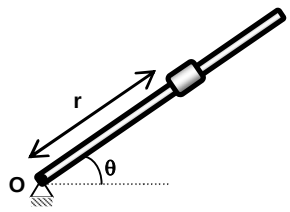
(A) For any value of m_1, m_2 and m_3 there can be at most 2 collisions.

(B) If $m_1 = 4m$, $m_2 = m$ and $m_3 = 3m$ then for any value of v_0 there will be 2 collisions.

(C) If $m_1 = 4m$, $m_2 = m$ and $m_3 = 3m$ then the final speed of m_3 will be 2.5 m/s.

(D) If $m_1 = 4m$, $m_2 = m$ and $m_3 = 3m$ then the final speed of m_2 will be 1.5 m/s.

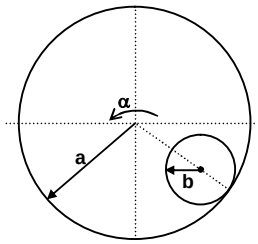
4. The rotation of rod OA about O is defined by the relation $\theta = \pi\left(4t^2 - 8t\right)$, where θ and t are expressed in radians and seconds, respectively. Collar B slides along the rod so that its distance from O is $r = 10 + 6\sin\pi t$, where r and t are expressed in metres and seconds, respectively. When t = 1 s,



(A) The speed of the collar in radial direction is 6π m/s.

(B) The speed of the collar in tangential direction is 80π m/s
(C) The speed of the collar in radial direction is zero.
(D) The speed of the collar in tangential direction is zero.

5. A perfectly rough cylindrical grindstone, of radius a = 1m, is rotating with a uniform angular acceleration about its axis, which is horizontal. If a sphere in contact with its edge can remain with its centre at rest, The angular acceleration of the grindstone (in rad/s^2) can be (take g = 10m/s^2) (There is no slipping between the sphere and the cylindrical grindstone)



(A) 15
(B) 20
(C) 25
(D) 30

6. A train is travelling at a speed of v_0. Then it decelerates uniformly during a time of t to finally stop. In one of the wagons there is a small object on the floor. The wagon long enough, so that the object does not collide with any of the walls of the wagon. ℓ is the distance covered by the object on the floor (with respect to train) , μ is the coefficient of friction between the floor of the wagon and the object.

(A) If $\frac{v_0}{t} < \mu g, \ell = 0$

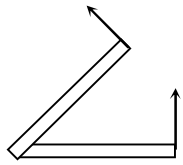
(B) If $\frac{v_0}{t} > \mu g, \ell = \frac{v_0^2}{4\mu g} - \frac{\mu g t^2}{4}$

(C) If $\frac{v_0}{t} > \mu g, \ell = 0$

(D) If $\frac{v_0}{t} > \mu g, \ell = \frac{v_0^2}{2\mu g} - \frac{v_0 t}{2}$

7. A particle is projected from a given point with a given velocity v = 10 m/s and after hitting a smooth vertical wall returns to the point from which it started. The distance of the initial point from the wall can be: (Take g = 10m/s^2) :
(A) 3 m
(B) 4 m
(C) 4.5 m
(D) 6 m

8. A long thin uniform rod lies flat on the table as shown. One end of the rod is slowly pulled up by a force that remains perpendicular to the rod at all times. The rod is to be brought to the vertical position without any slipping of the bottom end? The coefficient of static friction between rod and ground can be:



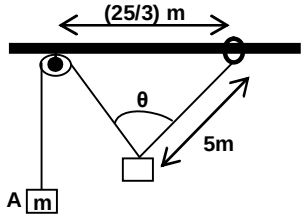
(A) 0.2
(B) 0.3
(C) 0.4
(D) 0.5

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

A mass of 1 kg is supported by a cord 5 m long fastened to a massless ring free to move on a horizontal rod , as shown in Figure. The coefficient of static friction between the ring and the rod is 0.75. A second cord is fastened to the weight and passes over a pulley fastened to the rod (25/3) m to the left of the ring. Block A of mass m is attached to the other end of this cord.



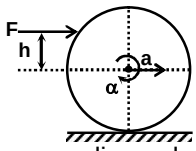
	List-I		List-II
(I)	Tension in the sting connected to block A in Newton .	(P)	6
(II)	Tension in the 5m cord in Newton.	(Q)	8
(III)	The value of θ is π/k radians , what can be the possible value/s of k.	(R)	2
(IV)	Acceleration of block A(in m/s^2).	(S)	0
		(T)	4
		(U)	12

9. If mass m is such that the ring just begins to slip. The correct match will be,
Options
(A) $I \rightarrow P, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$
(B) $I \rightarrow Q, II \rightarrow P, III \rightarrow T, IV \rightarrow S$
(C) $I \rightarrow T, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$
(D) $I \rightarrow U, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$

10. The value of m was initially chosen so that the ring just began to slide. Now , the ring is welded to the rod and mass of block A is increased to 5m , holding the 1kg mass . Immediately after the system is realeased then the correct match will be,
Options
(A) $I \rightarrow P, II \rightarrow Q, III \rightarrow R, IV \rightarrow T$
(B) $I \rightarrow Q, II \rightarrow P, III \rightarrow T, IV \rightarrow U$
(C) $I \rightarrow T, II \rightarrow U, III \rightarrow R, IV \rightarrow Q$
(D) $I \rightarrow U, II \rightarrow Q, III \rightarrow R, IV \rightarrow P$

Answer the following by appropriately matching the list based on the information given in the paragraph.

A solid body of mass m and radius 'R' is placed on a rough horizontal surface. When a horizontal force 'F' is applied to the body at a height 'h' above the centre level, the body starts rolling without slipping on the horizontal surface as shown.



List –I gives the four different shapes of the given solid body and corresponding values of F and h while List –II gives the magnitude of some quantities.

	List-I		List-II
(I)	For a ring, F = 2mg, $h = \frac{R}{2}$	(P)	1
(II)	For a disc, F = mg, $h = \frac{3R}{4}$	(Q)	$\frac{1}{3}$
(III)	For a hollow sphere, $F = \frac{5mg}{8}$, $h = \frac{R}{3}$	(R)	$\frac{7}{9}$
(IV)	For a solid sphere, $F = \frac{mg}{2}$, $h = \frac{3R}{5}$	(S)	$\frac{8}{21}$
		(T)	$\frac{1}{4}$
		(U)	$\frac{1}{7}$

11. If acceleration of the ring is 'a_0' then the acceleration of the different shapes of the body in a_0 units will be

Options

(A) $I \to P, II \to R, III \to Q, IV \to S$
(B) $I \to Q, II \to P, III \to S, IV \to U$
(C) $I \to P, II \to T, III \to R, IV \to S$
(D) $I \to T, II \to Q, III \to R, IV \to U$

12. If frictional force acting on the ring is 'f_0' then the frictional force acting on the different shapes of the body in 'f_0' units will be

Options

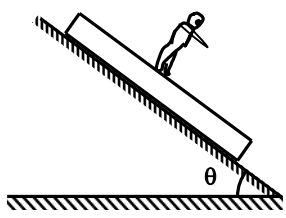
(A) $I \to P, II \to Q, III \to R, IV \to S$
(B) $I \to Q, II \to P, III \to S, IV \to U$
(C) $I \to P, II \to Q, III \to T, IV \to U$
(D) $I \to T, II \to Q, III \to R, IV \to U$

SECTION – C
(Numerical Answer Type)

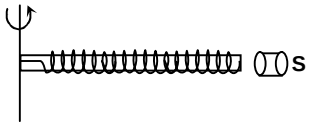
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A mass of m = 1kg moves initially with a velocity of $v_0 = 10\sqrt{15}$ m/s. A constant power P = 1000 Watt is applied so as to increase its velocity. Find the time (in sec) that elapses before the acceleration is reduced to half of its original value.

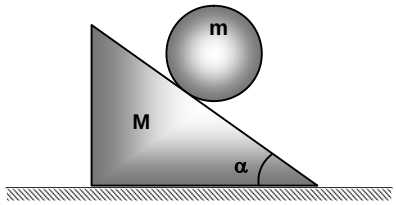
14. A plank, of mass M = 1kg and length ℓ = 7.2 m, is initially at rest along a line of greatest slope of a smooth plane inclined at an angle $\theta = 30°$ to the horizontal, and a man, of mass M_0 = 1kg, starting from the upper end walks down the plank so that it does not move. Find the time(in seconds) in which he gets to the other end of the plank .



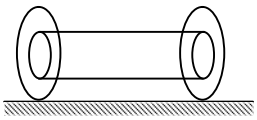
15. A rod of length ℓ = 2m is maintained to rotate with a constant angular velocity ω = 1 rad/s about a vertical axis passing through one end (fig). There is a spring of spring constant k = 1 N/m which just encloses rod inside it in natural length. One end of the spring is attached to axis of rotation. S is a sleeve of mass m = 0.4kg which can just fit on the rod. All surfaces are smooth. With what minimum kinetic energy (in J) sleeve should be projected so that it enters on the rod without impulse and completely compresses the spring.



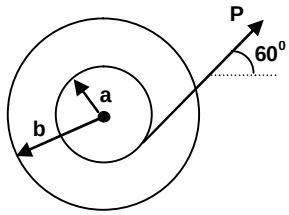
16. A wedge of mass M = 10 kg and inclination α = 30° is free to move on a smooth horizontal plane. A uniform solid cylinder of mass m = 20 kg is placed on the rough inclined face of the wedge. Find the acceleration (in m/s^2) of the centre of the cylinder relative to the wedge down the face. There is no slipping between the cylinder and the wedge.



17. Rope is coiled round a drum of radius a = 1.5 m. Two wheels each of radius b = 2 m are fitted to the ends of the drum, and the wheels and drum form a rigid body having a common axis. The system stands on the rough horizontal surface and a free end of the rope, after passing under the drum, is inclined at an angle of 60° to the horizontal. If a force P = 10 N be applied to the rope, find the magnitude of acceleration (in m/s^2) of centre of drum. Where mass of the system is 0.25 kg and its radius of gyration about the axis is k = 2 m.



Front view



Side view

18. At what speed (in m/s) must a pebble be thrown from a height of h = 3.6m, and at an angle of 30°, measured from the horizontal, if it is to hit the ground at an angle of 45°. (Air drag can be neglected.) (Take g = 10 m/s^2)

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which is/are correct regarding alkali metals:
   (A) Conductivities of alkali metal chloride in aqueous solution follows order LiCl < NaCl < KCl < RbCl
   (B) All alkali metal dissolved in liquid NH_3 their solubility follows order Li < Na < K
   (C) In the electrolysis of NaCl (aq) during formation of NaOH. Cl_2 (g) at cathode and H_2 (g) at anode are liberated
   (D) Negative value of ΔH_f^o (standard enthalpy of formation) LiF > NaF > KF > RbF > CsF

20. Correct statement are
   (A) In $\left[ICl_4\right]^-$ molecular ion, $Cl-I-Cl$ bond angle is 90°.
   (B) Bond dissociation energy follows order H – H > Cl – Cl > Br – Br
   (C) Extent of overlap follows order p – p > p – s > s – s
   (D) When ether is added to the mixture of LiCl, NaCl, KCl, it is KCl which will be extracted to the ether solution.

21. Which of the following is/are true?
   (A) Titration of H_3BO_3 with NaOH require *cis*-polyhydroxy alcohol to move the reaction in forward direction.
   (B) Aqueous solution of Borax will act as buffer solution.
   (C) Elemental B can be obtained by reducing the B_2O_3 with carbon.
   (D) $(BN)_x$ is obtained when NH_4Cl and $Na_2B_4O_7$ is reacted at red hot temperature.

22. Select the correct statement(s):
   (A) $4\pi r^2\psi^2$ vs r plotted for a H-orbital curve contains 3 maxima and 2 angular node then orbital will be 5d.
   (B) There will be 13 electrons in Cu atom having magnetic quantum number is equal to zero.
   (C) There will be minimum four atoms of hydrogen atom require during de-excitation from 4th shell to 1st shell to cover all distinct spectral line.
   (D) The value of $(n+\ell+m)$ of last electron in Mg atom is 5.

23. $N_2O_4 \rightleftharpoons 2NO_2$
   In above equilibrium mixture of N_2O_4 and NO_2 at 425 K, contains N_2O_4 at 4 atm and NO_2 at 8 atm. Now volume of vessel becomes double at the same temperature.
   (A) New pressure of N_2O_4 is 1.53 atm
   (B) New pressure NO_2 is 4.945 atm
   (C) Mole percentage of N_2O_4 is 23.63%
   (D) New pressure of NO_2 is 7.26 atm

24. Which of the following statements is/are correct?
   (A) Diamond is a covalent solid and co-ordination number of each carbon atom is 4.

(B) The CCP and HCP structure of a given element would have same co-ordination number and same packing fraction.
(C) On decreasing the pressure, coordination number of CsCl decreases to that of NaCl.
(D) In fluorite structure of CaF_2, Ca^{+2} ion occupy tetrahedral void and F^- ion at CCP point.

25. Which among the following statements are not correct?
(A) CaC_2 and Mg_2C_3 on hydrolysis gives alkyne.
(B) Be_2C and Al_4C_3 on hydrolysis gives alkene.
(C) Linear silicones are formed by hydrolysis of monoalkyl chloro silane.
(D) In pyrosilicate two oxygen atoms is shared between two $\left[SiO_4^{4-}\right]$ unit.

26. Which of the following are not true?
(A) A mixture of 2 mole of CH_3COONa and 1 mol of HCl will act as buffer solution.
(B) 200 ml of $\frac{M}{3}$ solution of H_3PO_2 is completely neutralized by 100 ml of 1M $Ca(OH)_2$ solution.
(C) pH of 0.5 M NaH_2PO_2 solution is 7.45 where pK_a of H_3PO_2 is 1.2.
(D) NaH_2PO_2 is a amphoteric salt.

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Match the following List-I with List-II:

LIST–I		LIST–II	
(I)	NaCl type (AB)	(P)	Co-ordination number 6 : 6
(II)	Antifluorite – (A_2B)	(Q)	Co-ordination number 4 : 8
(III)	Fluorite (AB_2)	(R)	Co-ordination number 8 : 4
(IV)	Diamond	(S)	Effective atom/ions in unit cell 8
		(T)	Effective atom/ions in unit cell 4
		(U)	Packing fraction 34%

27. Which of the following options has the correct combination considering List-I and List-II?
Options
(A) (IV), (S)
(B) (III), (Q)
(C) (II), (R)
(D) (I), (T)

Match the following List-I with List-II:

LIST–I		LIST–II	
(I)	NaCl type (AB)	(P)	Co-ordination 6 : 6
(II)	Antifluorite – (A_2B)	(Q)	Co-ordination 4 : 8
(III)	Fluorite (AB_2)	(R)	Co-ordination 8 : 4
(IV)	Diamond	(S)	Effective atom/ions in unit cell 8
		(T)	Effective atom/ions in unit cell 4
		(U)	Packing fraction 34%

28. Which of the following options has the correct combination considering List-I and List-II?
Options
(A) (II), (S)
(B) (I), (S)
(C) (IV), (Q)
(D) (III), (P)

Match the following List-I with List-II:

LIST–I		LIST–II	
(I)	KO_2	(P)	Paramagnetic
(II)	K_2O_2	(Q)	Diamagnetic
(III)	K_2O	(R)	Aqueous alkaline solution
(IV)	KO_3	(S)	CO_2 absorber and O_2 releaser
		(T)	Coloured
		(U)	Used as breathing mask in submarine

29. Which of the following options has the correct combination considering List-I and List-II?
Options
(A) (I), (Q)
(B) (II), (P)
(C) (I), (U)
(D) (IV), (S)

Match the following List-I with List-II:

LIST–I		LIST–II	
(I)	KO_2	(P)	Paramagnetic
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(III)	K_2O	(R)	Aqueous alkaline solution
(IV)	KO_3	(S)	CO_2 absorber and O_2 releaser
		(T)	Coloured
		(U)	Used as breathing mask in submarine

30. Which of the following options has the correct combination considering List-I and List-II?
Options
(A) (IV), (Q)
(B) (I), (T)
(C) (III), (S)
(D) (III), (P)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. Balance the chemical reaction given below:

$$x Mn(NO_3)_2 + yKOH + S \longrightarrow K_2MnO_4 + K_2SO_4 + NO + zH_2O$$

Find $\dfrac{x+y+z}{10}$

32. An inorganic mixture containing 0.5 mole of PCl_3 and 0.5 mol of PCl_5 was dissolved in excess of water. Find the number of moles of NaOH required to neutralized this solution.

33. RH_2 (ion exchange resin) can replace Ca^{+2} in water. If 1 litre of water is passed in ion exchange resin and pH of solution found to be 5.
Find hardness of water in ppm scale in term of Ca^{+2}.

34. One millimole of CO_2 is absorbed in 200 ml of 0.1 M NaOH. If the resulting solution is titrated with HCl using phenolphthalein as indicator, find millimole of HCl required when indicator changes it's colour.

35. For the reaction
$A(g) \rightleftharpoons B(g) + C(g)$
Average vapour density of mixture is 83.33. If molar mass of A, B and C are 100, 60 and 40 g. Find the number of moles of B formed at equilibrium starting with one mole of A.

36. 49 gram of a sample of oleum containing 20% free SO_3 was neutralized by $\frac{M}{10}NaOH$. The volume of NaOH in litre require for neutralization is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let f be a continuous function on [0, 1] such that $\int_x^1 f(t)dt \geq \frac{1-x^2}{2}$ $\forall$ x $\in$ [0, 1], then $\int_0^1 f^2(t)dt$ is always greater than or equal to

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{4}{19}$

(D) $\frac{3}{10}$

38. The number of function(s) f: [0, 1] $\to$ R such that $\int_0^1 \left(f\left(x^{1/k}\right)\right)^{n-k}dx = \frac{k}{n}$ $\forall$ k $\in$ {1, 2, 3, ..., n – 1}, where n is an odd integer greater than 1, is

(A) 1
(B) 2
(C) 5
(D) 7

39. Let f: R $\to$ R be a thrice differentiable function such that f(x), f′(x), f″(x) and f‴(x) are positive $\forall$ x $\in$ R. Moreover, f‴(x) < f(x) $\forall$ x $\in$ R, then

(A) f′(x) < 2f(x) $\forall$ x $\in$ R

(B) $f'(x) < \left(\frac{3}{2}\right)^{\frac{1}{3}} f(x)$ $\forall$ x $\in$ R

(C) $f'(x) < \frac{5}{2}f(x)$ $\forall$ x $\in$ R

(D) f′(x) < 3f(x) $\forall$ x $\in$ R

40. If y(x) be the solution of the differential equation $y \cdot y'' = 2x(y')^2$ given y′(0) = –4, y(0) = 1, then

(A) $y(1) = e^4$
(B) $y(2) = e^{8/3}$
(C) $y(x) \in (0, \infty) - \{e^2\}$
(D) $y(x) \in (0, \infty) - \{e^3\}$

41. Let f: [0, 1] $\to$ R be continuously differentiable function and $\left|\frac{f(0)+f(1)}{2} - \int_0^1 f(x)dx\right| \leq \alpha \max_{x \in [0,1]} |f'(x)|$, then α can be

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{3}{10}$

(D) 1

42. If $\frac{e^x + e^{-x}}{2} \le e^{cx^2} \quad \forall\, x \in R$, then c can be

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{e}$

(D) 1

43. Which of the following are INCORRECT?

(A) $f(x) = \{\sin x\} \cos x (\cos^{-1} \cos x)^3$ is not differentiable at $2n\pi$, $\left(2n\pi + \frac{\pi}{2}\right)$ and $(2n + 1)\pi$, $n \in Z$ (where {.} represents fractional part function)

(B) $f(x) = |(x - 1)(x - 2)(x - 3)||\sin \pi x|$ is not differentiable at 3 points in $x \in [0, 4]$

(C) $f(x) = ||x - 2| - |x - 8|| - 3|x - 5|^3 + 8x + 1 + \cos(|x - 8|)$ not differentiable at 3 points

(D) $f(x) = (x^2 - 3x + 2)^{3/5} \sin(x^2 - 2x)$ is not differentiable at x = 1

44. Let f: [0, 1] $\to$ R be a continuous function, differentiable on (0, 1) such that $\int_0^a f(x)dx = 0$ for some $a \in (0, 1)$, then

(A) $\left|\int_0^a f(x)dx\right| \le \frac{1-a}{2} \max_{0<x<1} |f'(x)|$

(B) $\left|\int_0^1 f(x)dx\right| > \frac{1-a}{2} \max_{0<x<1} |f'(x)|$

(C) $\left|\int_0^1 f(x)dx\right| = \frac{1-a}{2} \max_{0<x<1} |f'(x)|$ for some function f(x)

(D) none of the other options is correct

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

MM-I. Let $f_1(x) = \lim_{n\to\infty} (x-1)\frac{(x^{4n}-1)}{(x^{4n}+1)}$;

$f_2(x) = \{x\} \cos^2\left(\frac{2x-1}{2}\right)\pi$; (where {.} denotes fractional part function x);

$$f_3(x)=\begin{cases} x^2-x\sqrt{3} & ; \quad x \text{ is irrational} \\ 0 & ; \quad x \text{ is rational} \end{cases};$$

$$f_4(x)=\begin{cases} 0 & ; \quad x \text{ is irrational} \\ \frac{1}{n} & ; \quad x \text{ is rational number of the form } x=\frac{m}{n} \text{ in lowest terms with } m, n>0 \end{cases}$$

LIST–I		LIST–II	
(I)	$f_1(x)$ is discontinuous at	(P)	–1
(II)	$f_2(x)$ is discontinuous at	(Q)	0
(III)	$f_3(x)$ is discontinuous at	(R)	1
(IV)	$f_4(x)$ is discontinuous at	(S)	2
		(T)	$\sqrt{2}$
		(U)	$\sqrt{3}$

45. Which of the following is the only INCORRECT combination?
(A) (I), (P)
(B) (IV), (S)
(C) (IV), (R)
(D) (IV), (T)

46. Which of the following is the only CORRECT combination?
(A) (II), (P)
(B) (III), (T)
(C) (III), (Q)
(D) (IV), (U)

MM-II. List-I represents equations and List-II represents number of positive real roots of the corresponding equations

LIST–I		LIST–II	
(I)	$(x^3-x+1)^3-x^3+x=0$	(P)	0
(II)	$x^6-x^4+2x^3-x^2+1=0$	(Q)	1
(III)	$\frac{1}{x-1}+\frac{1}{x-2}+\frac{1}{x-3}-x^{\frac{3}{5}}=0$	(R)	2
(IV)	$x^4-2^x=0$	(S)	3
		(T)	4
		(U)	5

47. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (I), (T)
(C) (III), (T)
(D) (III), (R)

48. Which of the following is the only CORRECT combination?
(A) (IV), (R)
(B) (IV), (Q)
(C) (II), (Q)
(D) (II), (R)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let f(x) be a differentiable function with a continuous derivative $\forall\ x \in R$. If f(1) = 2 and f(2) = 5 and f′(x) takes only rational values, then f(3) + f′(0) is equal to

50. If f(x) is a function with a continuous third derivative on [0, 2] such that f(0) = f′(0) = f″(0) = f′(2) = f″(2) = 0 and f(2) = 2, then minimum number of value(s) of $c \in [0, 2]$ such that $f'''(c) \geq 6$ is

51. Let f: $(0, \infty) \to R$ be a twice differentiable function such that $|f''(x) + 2xf'(x) + (x^2 + 1)f(x)| \leq 2$ $\forall\ x \in R$ and $\lim\limits_{x\to\infty} (xf(x) + f'(x)) = 5$, then $\lim\limits_{x\to\infty} f(x)$ is equal to

52. Let f: $R \to R$ be a continuous function $\forall\ x \in R$ such that $f(x) = f\left(x^2 + \frac{1}{4}\right)$ and f′(0) = 0, then $f'\left(\frac{1}{4}\right)$ is equal to

53. Let f(x) be a three times differentiable function (defined on R) such that f(x) has at least five distinct real zeroes, then number of distinct real zeroes of f(x) + 6f′(x) + 12f″(x) + 8f‴(x) is/are

54. $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \left(\frac{k}{n^2}\right)^{\frac{k}{n^2}+1}$ is equal to